

Genetic mismatches with vaccine strains facilitate the persistence of influenza viruses during the COVID-19 pandemic

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Introduction

- Seasonal influenza causes highly transmissible respiratory disease resulting in significant morbidity and mortality in worldwide, with H1N1pdm09 and H3N2 as a major circulating influenza A virus subtypes.
- During the COVID-19 pandemic, Non-pharmaceutical interventions (NPI) led to a dramatic reduction in influenza cases.
- However, as these measures were lifted, influenza re-emerged with altered patterns of clade predominance.
- Vaccination is crucial strategy for preventing seasonal influenza, and it also shapes selective pressure that influence future outbreaks.
- The evolutionary dynamic of post-pandemic influenza viruses has not been thoroughly investigated.
- In this study, we analyzed their vaccine escape potential to evaluate the role of vaccination as a selective pressure shaping influenza evolution during the COVID-19 pandemic.

Materials & Methods

- Preparing sequence data set**
 - The HA sequences of seasonal influenza H1N1 pdm09 and H3N2 were retrieved from the GISAID (July 1, 2016-Jun 31, 2023).
 - Each sequence was quality-controlled and assigned to a clade using Nextclade (<https://clades.nextstrain.org>).
 - To reduce redundancy, sequences were clustered using CD-HIT on a per-season and per-region basis.
- Vaccine mismatch & antigenic drift analysis**
 - To evaluate antigenic mismatches, HA amino acid sequences from the selected circulating sequences were compared with vaccine strains that used in pre-pandemic and COVID-19 pandemic period (2016-2023).
 - Antigenic mismatches were counted at each antigenic site (A–E).
 - Statistical significance of temporal trends in vaccine mismatch patterns was assessed using one-way ANOVA.
 - For clades that circulated during the pandemic period, selection pressure was examined using HyPhy, focusing on the identification of positively selected sites.
 - Additional antigenic characterization was performed using the sequence-based antigenic cartography method.
 - Vaccine efficacy (VE) was predicted for influenza strains belong to pandemic-dominant clades but isolated in pre-pandemic period using $P_{epitope}$ model.
 - An AD matrix was constructed by calculating and averaging the five epitopic distances (EDs) between each virus with the Hamming distance.
- Phylogenetic analysis**
 - To ensure accurate clade classification, time-scaled maximum clade credibility (MCC) phylogenetic trees were constructed.
 - A fixed clock model was employed to examine changes in substitution rates between the pre-pandemic and the pandemic periods. For H1N1pdm09, clade 6B.1A.5a.2 and its subclades were constrained as monophyletic, while for H3N2, clade 3C.2a1b.2a.1 and 3C.2a1b.2a.2 and its subclades and clade 3C.2a1b.2a.2 were constrained as monophyletic in the substitution analyses.
 - Viral population dynamic were estimated using GMRF Skygrid model.
 - Ancestral sequence reconstruction (ASR) was performed using TreeTime to infer amino acid substitutions along the phylogeny. Substitutions identified by ASR were annotated at major branching points of predominant clades to highlight lineage-defining changes.

Result & Discussion

- Result**
- During the COVID-19 pandemic, seasonal influenza activity was markedly reduced in both H1N1pdm09 and H3N2 (**Figure 1**).
 - The outbreak decline was followed by notable shifts in the dominant clades; in H1N1pdm09, the 6B.1A.5a.2 subclades, and in H3N2, the 3C.2a1b.2a.1 and 3C.2a1b.2a.2 subclades become dominant (**Figure 2**).
 - In both H1N1pdm09 and H3N2, clades dominant in pandemic period showed higher mismatch with vaccine used in pre-pandemic period (**Figure 2, 3A, 3B**).
 - Statistical analysis showed antigenic differences between pre-pandemic vaccine × pre-pandemic dominant clade and pre-pandemic vaccine × pandemic-dominant clade (**Figure 3C, 3D**).
 - Although the overall dN/dS ratios of both H1N1 and H3N2 lineages were below 1, indicating dominant purifying selection, most positively selected sites (H1N1: 3 /5, H3N2: 5 /7) were located within antigenic sites (**Table 1**).
 - The computed vaccine efficacy of strains that belong to pandemic-dominant clades but isolated in pre-pandemic period were lower than 41% for VE (53%) and 77% for VE (100%) (**Table 2**).
 - In both H1N1 and H3N2, annual antigenic drift corresponded with vaccine updates.
 - Antigenic drift of both influenza lineage was observed in the direction away from the vaccine strains used in the same period (**Figure 5A, B**).
 - During the early pandemic phase, both H1N1pdm09 and H3N2 exhibited reduction in viral population size (**Figure 5A, B**).
 - Phylogenetic tree-based analysis of amino acid substitutions showed that substitutions occurred more frequently at branching points of pandemic-clades than at those of pre-pandemic clades (**Figure 5A, 5B**).

Discussion

- The COVID pandemic influenced both the intensity of influenza circulation and the evolutionary dynamics of circulating lineages.
- Pandemic-dominant viruses showed statistically significant antigenic mismatch with pre-pandemic vaccines, supported by selection signal at antigenic site and reduced predicted vaccine efficacy.
- This suggests that vaccine-induced immune pressure during the COVID-19 pandemic likely contributed to clade conversion by driving adaptive changes at antigenic epitopes.
- Antigenic distance analysis farther indicate that vaccine pressure contribute to antigenic drift in both H1N1pdm09 and H3N2.
- The reduction in seasonal influenza population size during the pandemic was accompanied by selective pressures that facilitated the replacement of viral clades.
- Collectively, these findings indicate that the pattern of antigenic change during the COVID-19 pandemic differed from that observed in the pre-pandemic period.
- Our results demonstrate that the COVID-19 pandemic not only reduced influenza circulation but also altered its evolutionary trajectory.
- These findings suggest that not only mitigation policies during pandemic period but also vaccination appear to have contributed to viral selection during COVID-19 pandemic.

Result

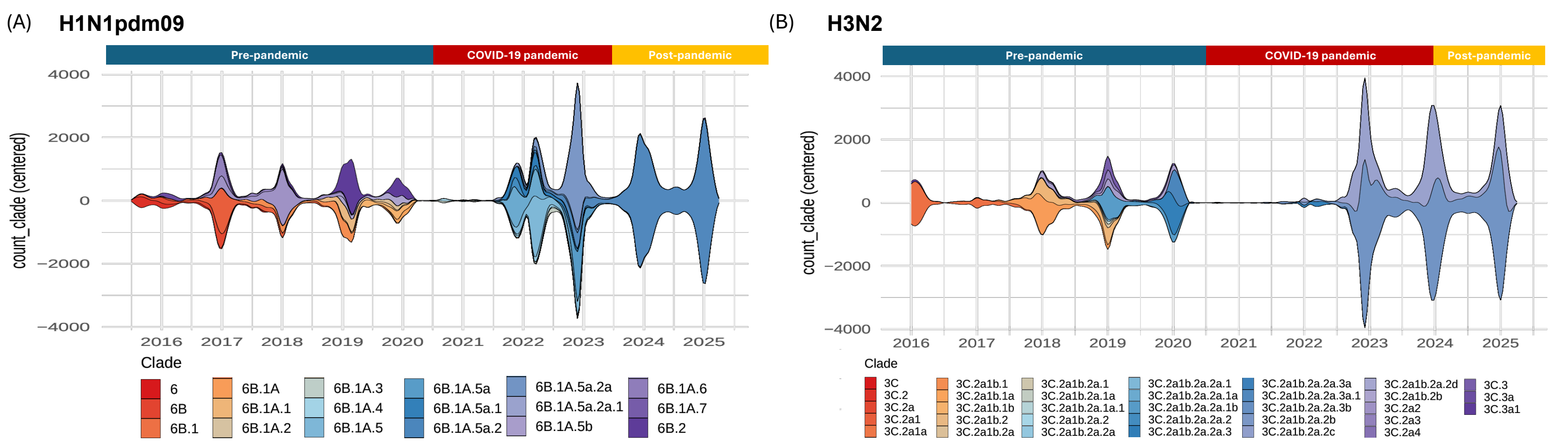


Figure 1. Temporal changes in influenza A clades (H1N1pdm09 and H3N2) before, during and after the COVID-19 pandemic. Stream plots illustrate the relative abundance and diversity of circulating clades over time. (A) H1N1pdm09 and (B) H3N2 sequences are shown as stream plots, with each clade represented by a distinct color.

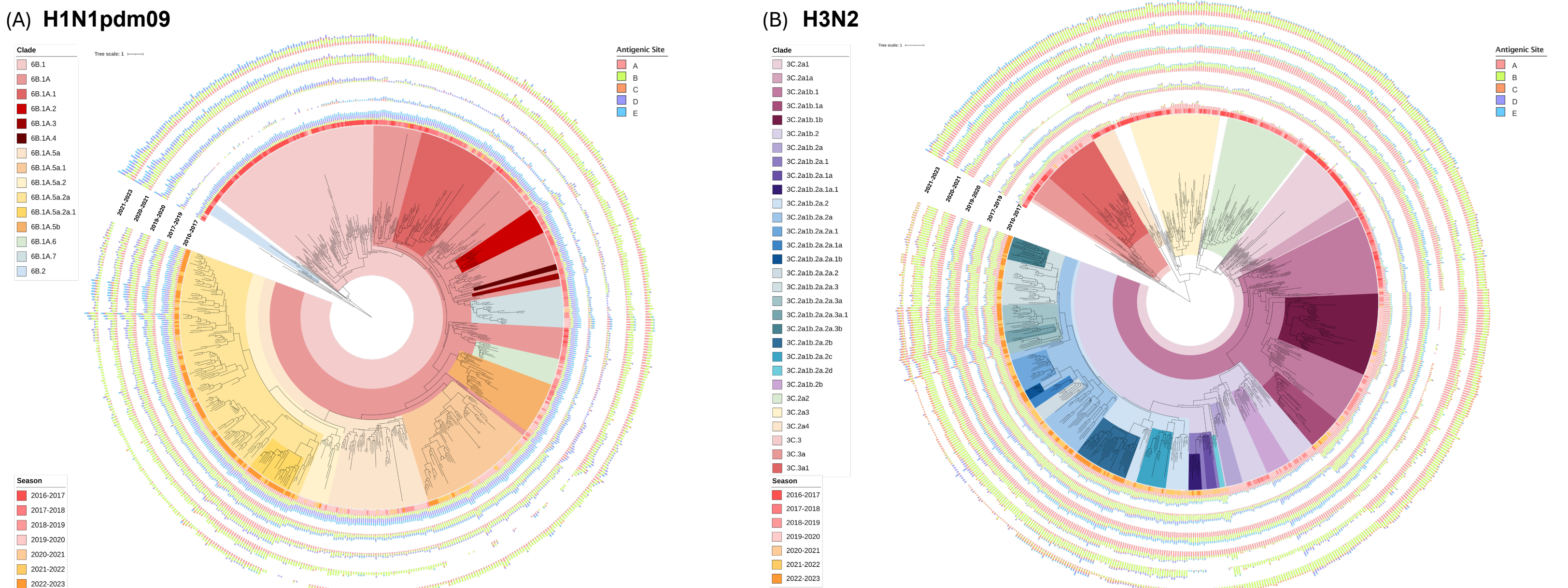


Figure 2. Time-scaled maximum clade credibility (MCC) phylogenetic tree of influenza A viruses (H1N1pdm09 and H3N2). Major circulating clades are highlighted in distinct colors on the tree structure, and concentric rings represent influenza seasons from 2016–2017 to 2022–2023. The outermost ring indicates the number of antigenic mismatches with the corresponding seasonal vaccine, with epitope sites (A–E) shown in distinct colors.

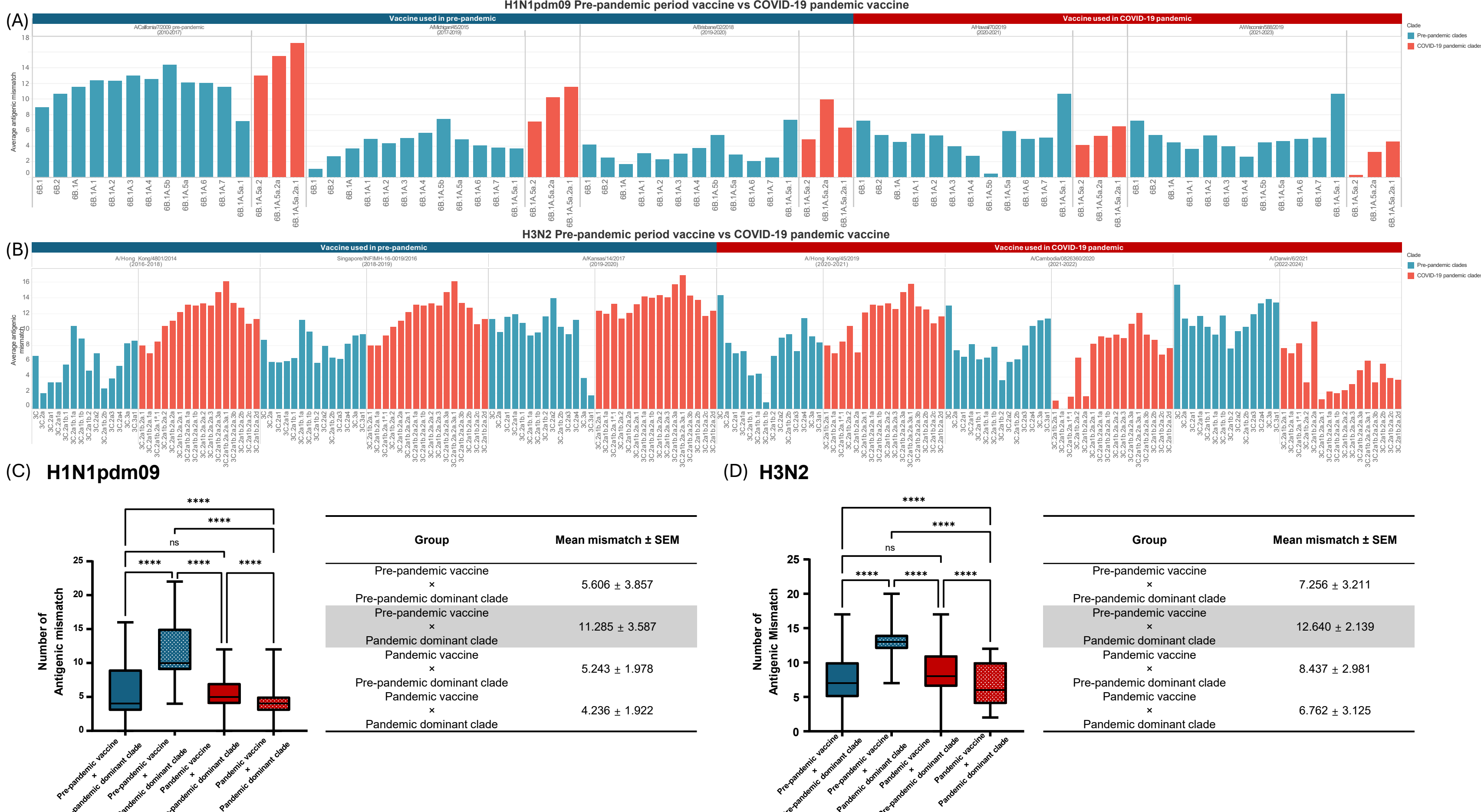


Figure 3. Antigenic mismatch patterns and one-way ANOVA results for H1N1pdm09 and H3N2 hemagglutinin sequences. (A) H1N1pdm09 and (B) H3N2 antigenic site mismatches between circulating clades and vaccine strains before and during the COVID-19 pandemic. Stacked bar plots show the mean number of amino acid mismatches at defined antigenic sites (A–E) for pre-pandemic clades versus pandemic-dominant clades against corresponding vaccine strains. Bar graphs in (C) and (D) summarize mean mismatch and standard error of the mean (SEM).

Table 1. Sites under selection detected by HyPhy analyses across antigenic and non-antigenic regions of COVID-pandemic period dominant clade HA. This table summarizes codon sites in hemagglutinin (HA) identified as being under positive selection by HyPhy methods (MEME, FEL, SLAC, and FUBAR). Sites are categorized according to defined antigenic sites (A–E) or non-antigenic regions. Codons listed were supported by at least one method ($p < 0.1$ or posterior probability > 0.9).

Lineage	dN/dS	Methods of positive selection								Total number of site	The number of positive selected site in epitope
		Position	MEME Antigenic site	Position	FEL Antigenic site	Position	SLAC Antigenic site	Position	FUBAR Antigenic site		
H1N1pdm09	0.2381	70	E	70	E	-	-	-	-	5	3
		113	Non-antigenic site	-	-	-	-	-	-		
		187	B	-	-	-	-	-	-		
		270	C	-	-	-	-	-	-		
		279	Non-antigenic site	-	-	-	-	-	-		
H3N2	0.2551	48	-	135	A	-	135	A	-	7	5
		53	C	-	-	-	-	124	A		
		135	A	-	-	-	-	135	A		
		183	Non-antigenic site	-	-	-	-	347	Non-antigenic site		
		347	Non-antigenic site	-	-	-	-	-	-		

Table 2. Predicted vaccine efficacy (VE) for influenza strains belong to pandemic-dominant clades but isolated in pre-pandemic period. Vaccine efficacy was estimated using the $P_{epitope}$ model. VE(53%) was computed as $E = (-1.19 \times P_{epitope} + 0.53) \times 100$, and standardized VE(100%) was derived as $VE(100\%) = VE(53\%) / 0.53 \times 100$.

lineage	Vaccine strain	Vaccinated Season	Clade	Dominant Epitope	Number of Mismatch	Number of strains	$P_{epitope}$	Vaccine Efficacy(VE) 53%	100%
H1N1pdm09	A/Michigan/45/2015	2019-2019	6B.1A.5a.2	B	3	1	0.158	34.211	64.548
	A/Brisbane/02/2018	2019-2020	6B.1A.5a.2	B	2	11	0.105	40.474	76.365
H3N2	A/Kansas/14/2017	2019-2020	3C.2a1b.2a.2a	B	4	1	0.210	27.947	52.731

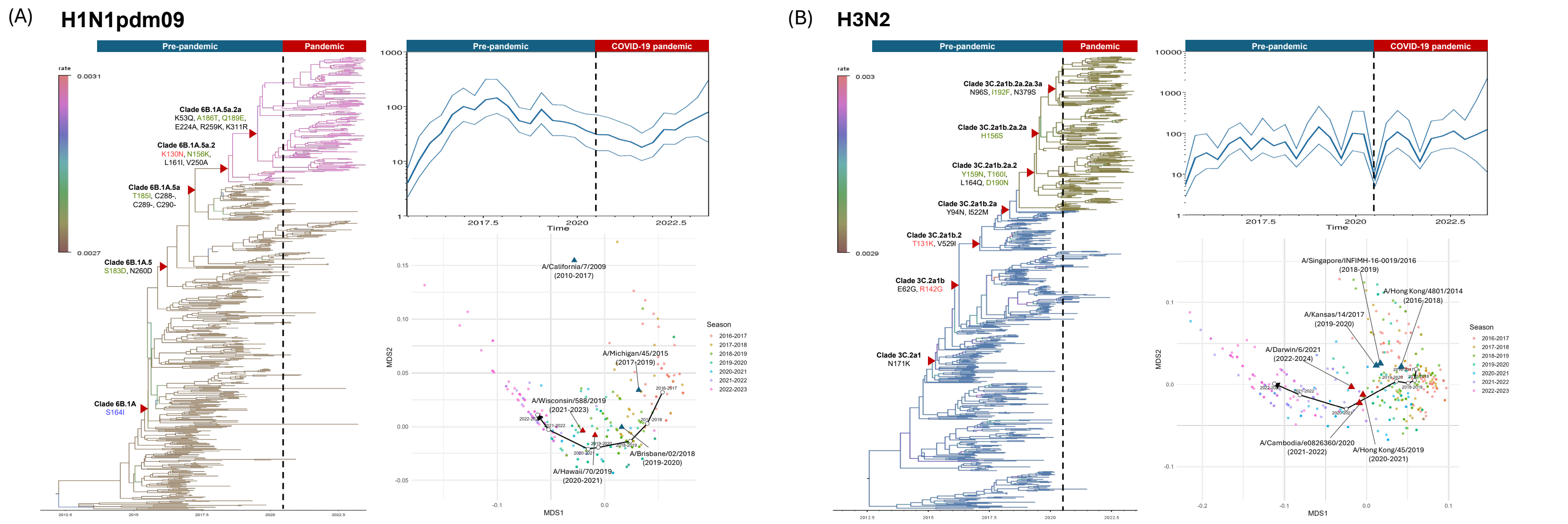


Figure 5. Phylogenetic analysis and sequence-based antigenic map of influenza A H1N1pdm09 and H3N2 viruses before and after COVID pandemic. (A) H1N1pdm09 and (B) H3N2. Branches of time-scaled maximum clade credibility (MCC) tree are colored according to substitution rates, and major substitutions are annotated at key clade branching points. Viral population dynamics estimated using the GMRF Skygrid model. Two-dimensional antigenic maps were generated by multidimensional scaling (MDS). The MDS plot presents virus strains (points), vaccine strains (triangles) and the seasonal trajectories of antigenic drift (black arrow).